Ideation Phase

Date	31 January 2025
Team ID	
Project Name	Visualization-Tool-For-Electric-Vehicle-Charge-And-Range-Analysis
Maximum Marks	4 Marks

Project Name: Visualization Tool for Electric Vehicle Charge and Range Analysis
Date: 31 January 2025
Team Members:
- Dixit Pandya (Team Lead)
- Shiv Pandya
- Nikhilkumar Raval
- Viraj Raval

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Problem Statement:
With the increasing adoption of Electric Vehicles (EVs), there is a growing need to analyze EV charging infrastructure, vehicle efficiency, price, and driving range. However, the data available is complex and spread across multiple datasets. Users such as city planners, fleet managers, and consumers require an easy-to-understand visualization platform to analyze EV charging patterns, battery performance, and vehicle efficiency.

Our team decided to build a Tableau-based visualization system integrated with a Flask web application that allows users to explore EV data interactively through dashboards and stories.

Step-2: Brainstorm, Idea Listing and Grouping

Idea 1: EV Charging Station Analysis
Analyze charging station locations across India and visualize them using geographical maps.

Idea 2: EV Model Comparison
Compare electric vehicles based on price, range, efficiency, and performance.

Idea 3: Battery Performance and Driving Range
Analyze battery efficiency and how it impacts driving range and charging cycles.

Idea 4: Consumer Decision Dashboard
Provide insights that help buyers compare EV models before purchasing.

Idea 5: EV Market Insights
Visualize market trends such as most efficient EV brands, number of models per brand, and body style distribution.

Step-3: Idea Prioritization

After evaluating all the brainstormed ideas, the team prioritized the following key features for the final project:

1. Charging Stations Map of India
2. Charging Stations by Region and Type
3. Top Speed Analysis by EV Brand
4. Price Comparison of EV Cars
5. Top 10 Most Efficient EV Brands
6. Body Style Distribution by Brand
7. Number of Models by Each Brand
8. EV Cars Available in India
9. Brand Filtering based on PowerTrain
10. Summary KPIs for Global EV Market
11. Summary KPIs for Indian EV Market

These visualizations were implemented using Tableau dashboards and stories.